



Case Study: Fashion E-Commerce Digital Marketing Analysis

Dsarea Data Analytics & Visualization

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A highly adaptable Biomedical Engineering graduate from Universitas Indonesia with strong analytical skills and solid problem-solving abilities. Experienced in turning raw data into clear, actionable insights using Excel, SQL, and Looker Studio. A fast learner with a firm belief in the value of data for informed decision-making and sustainable growth, committed to continuous learning and delivering effective data-driven solutions.



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Courses & Certifications



**Data Analytics & Visualization
Bootcamp**

| [Link Certificate](#)

(November, 2025)



Data Science & Data Analysis

| [Link Certificate](#)

(October, 2025)



Machine Learning with Python

| [Link Certificate](#)

(June, 2024)



Case Study: Background

This month, the **Conversion Rate (CR)** on the website and digital platforms of XYZ, a fashion e-commerce company, **declined by 2%** compared to the previous month. This drop **raises concerns as it may reduce revenue and weaken the effectiveness of ongoing marketing campaigns.**

At the request of the executive team, the marketing team needs to **analyze the root causes** of this decline and design strategies to **increase the conversion rate by 3% within the next three months**, focusing on website user experience optimization, social media engagement, and the effectiveness of digital advertising.





Main Objective Questions

- 1 Which marketing factors or channels contributed most to the decline in Conversion Rate (CR)?
- 2 How does user behavior on the website and social media influence their purchase decisions?
- 3 What strategies or optimizations can be implemented to increase the CR in the next three months?

Objectives

- 1 Increase the Conversion Rate (CR) by 3% within three months.
- 2 Identify the key factors and marketing channels influencing CR.
- 3 Understand visitor behavior across the website and social media platforms.



Data Definition

Column Name	Definition
User_ID	Unique ID of the visitor
Location	City or province of the visitor
Visit	Number of visits to the website
Session_Duration	Duration of the website session (in minutes)
Product_Category	Product category viewed



Data Definition

Column Name	Definition
Product_Page_Views	Number of product pages viewed
Ads_Name	Name of the advertisement or ad campaign
Transaction_date	Date the customer made a purchase
Impressions	Number of ad impressions
Clicks	Number of ad clicks



Data Definition

Column Name	Definition
Engagement	Number of interactions (likes + comments + shares)
Purchase	Purchase status (Yes/No)
Transaction_Amount	Transaction value (in currency)
Repeat_Customer	Returning customer status (Yes/No)

- **Awareness:** Impressions, Reach Rate
- **Interest:** Session_Duration, Engagement, CTR
- **Desire:** Product_Page_Views
- **Action:** Conversion_Rate, Transaction_Amount (AOV)

Data Sanity Check

Q1: Are there any duplicates or missing values?

```
# Check the number of duplicate rows
print("Number of duplicate rows:", df.duplicated().sum())

# View duplicate rows if any exist
df.duplicated()

Number of duplicate rows: 0
```

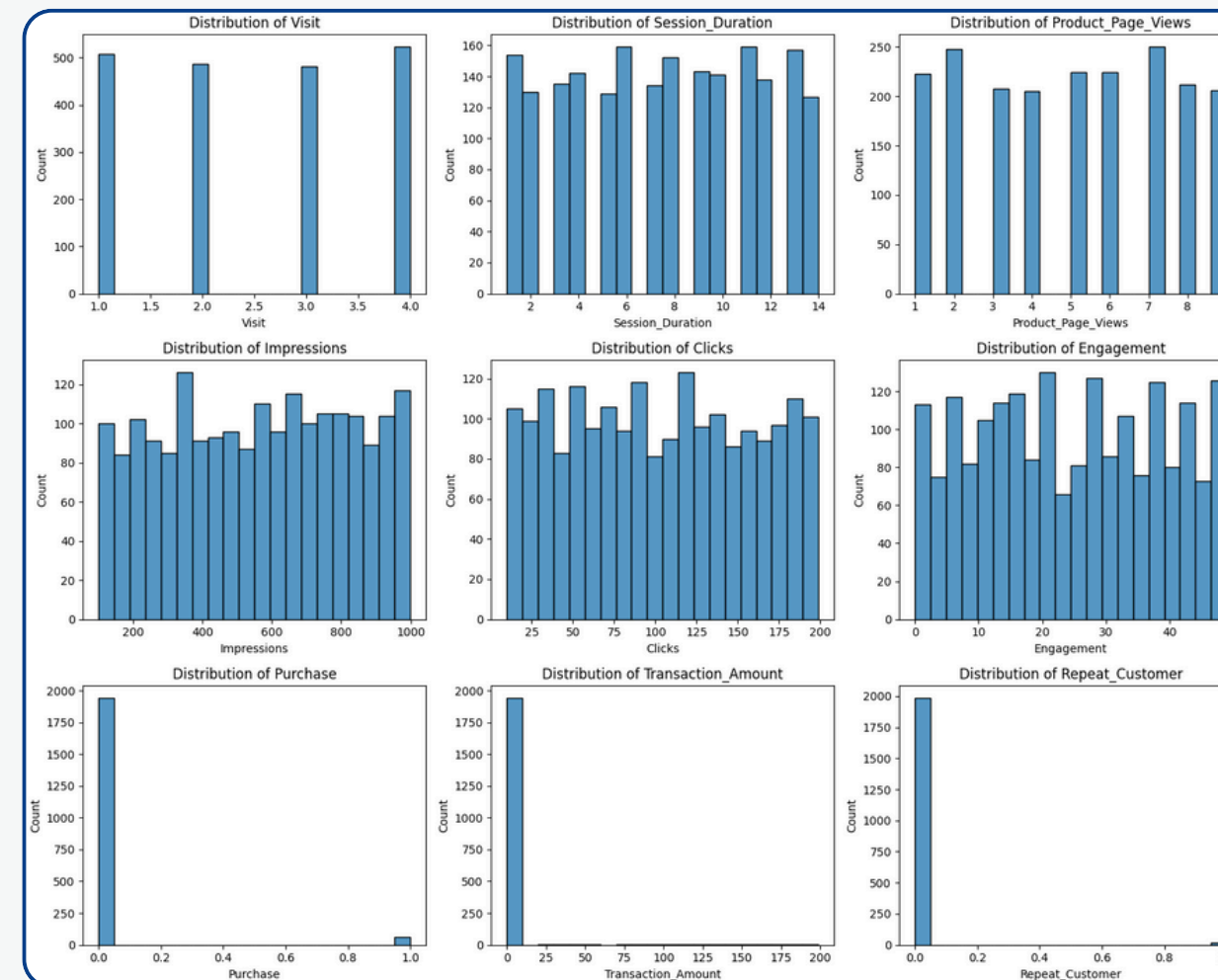
NO DUPLICATE

```
# Count missing values per column
print("Number of duplicate rows:", df.isnull().sum())

Number of duplicate rows: Date      0
User_ID      0
Location      0
Visit      0
Session_Duration  0
Product_Category  0
Product_Page_Views  0
Ads_Name      0
Impressions      0
Clicks      0
Engagement      0
Purchase      0
Transaction_Amount  0
Repeat_Customer  0
dtype: int64
```

NO MISSING VALUES

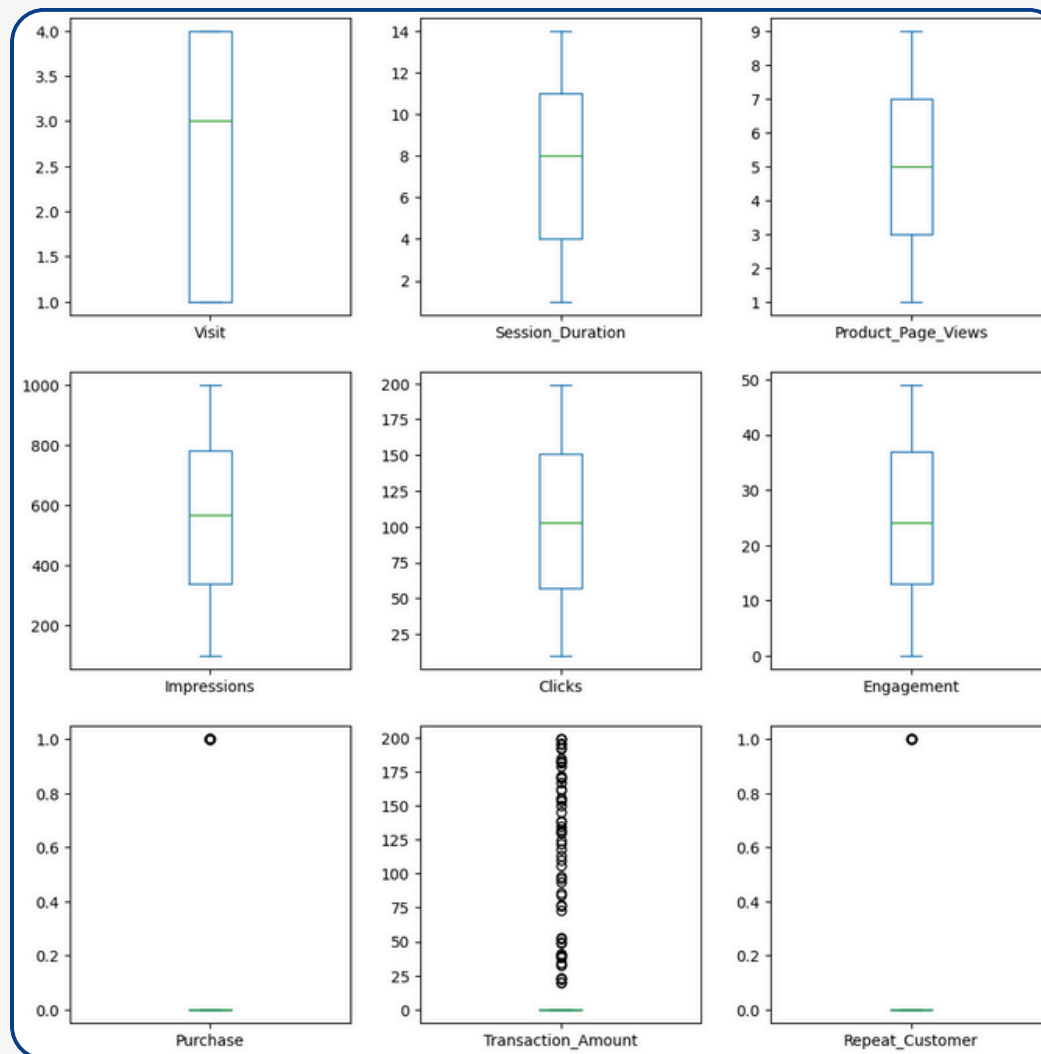
Q2: Are there any unreasonable data distributions?



Tools: Python

- **The distribution of key behavioral metrics** (Visit, Session Duration, Product Page Views, Impressions, Clicks, and Engagement) **appears stable and well-balanced across realistic value ranges**, indicating consistent data capture with no visible anomalies such as extreme spikes, gaps, or distorted patterns.
- **Purchase and Repeat_Customer appropriately show a strong dominance of non-conversion values (0)**, which aligns with typical e-commerce behavior where only a **smaller portion of users actually convert or return**.
- Meanwhile, **Transaction_Amount exhibits a right-skewed pattern**, reflecting the natural tendency for most transactions to occur at lower values, with fewer high-value purchases.
- Overall, the dataset is ready for the next analytical stage.

Data Sanity Check



Q3: Are there any outliers?

- The boxplots for the **main behavioral metrics** (Session Duration, Product Page Views, Impressions, Clicks, and Engagement) **show no visible outliers**.
- **Transaction_Amount is the only variable that displays several high-end outliers. This pattern is expected in an e-commerce context**, where a small proportion of customers accounts for disproportionately high transaction values. **These extreme observations are therefore informative** rather than problematic, as they may be linked to high-impact campaigns, premium customer segments, or bulk purchases. Removing them would risk eliminating meaningful business insights.

Tools: Python

- The **binary variables (Purchase and Repeat_Customer)** show a **strong class imbalance dominated by the value 0**. This distribution **accurately reflects a realistic business/marketing funnel**, in which **most visitors do not convert or become repeat customers**. The isolated values of 1 in the boxplot are not anomalies but represent successful conversion and retention events, making these variables critical indicators of campaign effectiveness and overall customer journey performance.

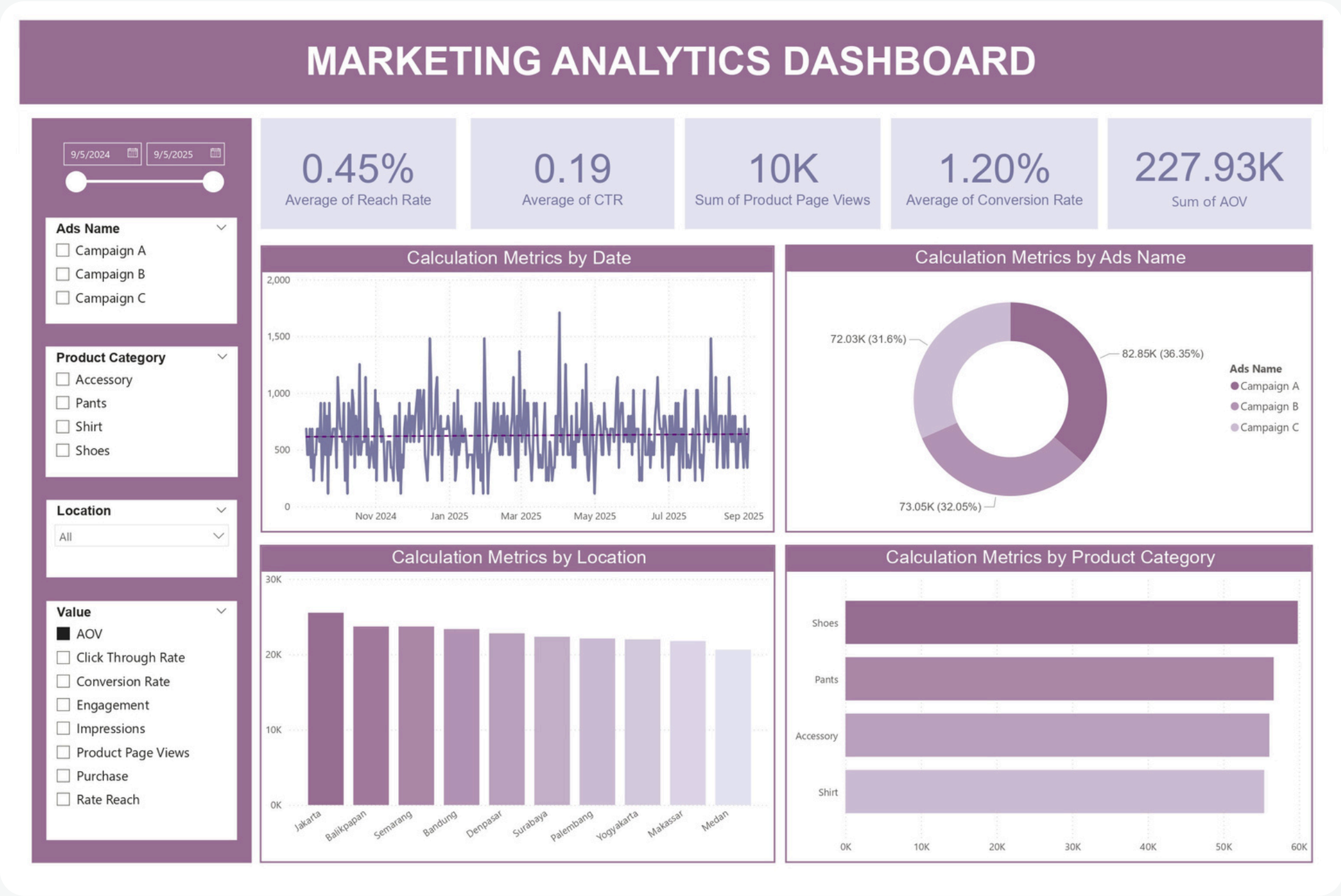
Dashboard Layout Design

Stakeholder: Executive Management

Filtering	HIGH LEVEL INFORMATION	Level 1
	Performance Trend & Campaign Share	Level 2
	Regional & Product Insights	Level 3

Marketing Analytics Dashboard

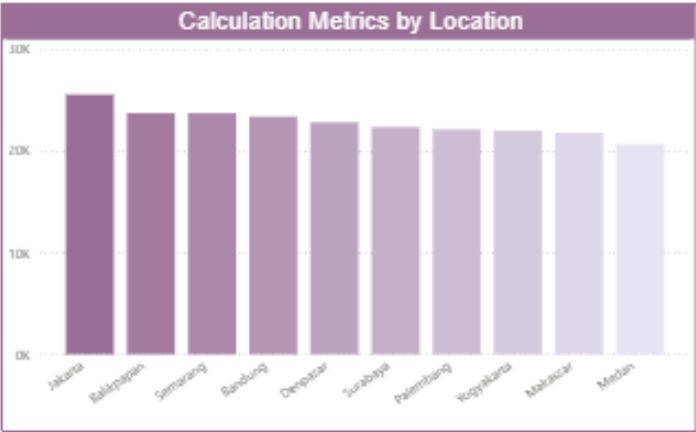
Tools: Power BI



Q1: Which marketing factors or channels contributed most to the decline in Conversion Rate (CR)?

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1 Deep Analysis – Location



Location	Reach Rate	Engage-ment	CTR	Impres-sion	Product Page View	Purchase	AOV	Conver. Rate
Balikpapan	0.44	24.35	17.95	5.06K	4.68	208	23.70K	0.78
Bandung	0.45	24.18	18.81	4.96K	4.68	205	23.36K	0.19
Jakarta	0.46	24.37	18.99	5.46K	4.88	224	25.53K	1.56
Makassar	0.42	25.30	18.13	4.83K	5.13	191	21.76K	1.50
Medan	0.42	23.70	18.31	4.29K	5.19	181	20.67K	1.39
Palembang	0.45	24.35	19.14	4.72K	5.12	194	22.10K	1.84

Location	Reach Rate	Engage-ment	CTR	Impres-sion	Product Page View	Purchase	AOV	Conver. Rate
Semarang	0.49	26.15	18.47	5.44K	5.16	208	23.70K	0.57
Surabaya	0.46	25.23	19.22	4.95K	5.11	196	22.33K	2.17
Yogyakarta	0.47	24.24	18.66	4.68K	4.70	193	21.99K	1.25
Denpasar	0.43	22.82	17.69	4.56K	5.04	200	22.79K	0.77

- **Reach Rate:** The averages in Denpasar (0.43), Makassar (0.42), and Medan (0.42) are the lowest.
- **Engagement:** Denpasar's average (22.82) is the lowest compared to the other cities (24).
- **CTR:** The averages in Balikpapan (17.95%) and Denpasar (17.69%) are the lowest.
- **Impression:** Medan (4.29k) has the lowest number

- **Product Page View:** The averages in Yogyakarta (4.70), Balikpapan (4.68), and Bandung (4.68) are the lowest.
- **Purchase:** Medan (181) has the lowest number.
- **AOV:** Medan (20.67K) is the lowest.
- **Conversion Rate:** The averages in Balikpapan (0.78%), Denpasar (0.77%), Semarang (0.57%), and Bandung (0.19%) are the lowest.

Summary

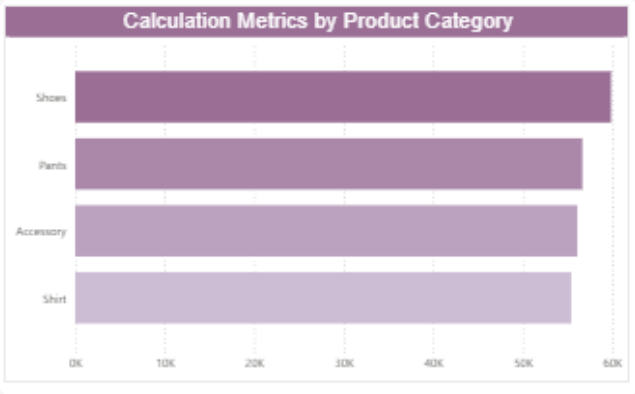
Several cities show low performance across almost all metrics, particularly Medan (Reach Rate, Impressions, Purchase, and AOV) and Denpasar (Reach Rate, Engagement, CTR, and Conversion Rate). Other cities, such as Bandung and Balikpapan, also show low scores in Product Page Views and Conversion Rate.

Recommendation

- **Makassar and Medan:** Run bundling and discount offers to increase purchase intent, launch awareness campaigns to drive initial traffic, provide product education (such as short-form videos) to encourage clicks to product pages, and activate seasonal promotions to accelerate purchases.
- **Denpasar:** Increase engagement through creative content and loyalty programs. You may also implement bundling, discounts, and awareness campaigns to broaden traffic reach.
- **Bandung, Balikpapan, Yogyakarta:** Optimize product pages (images, descriptions, ratings, trust badges), offer limited-time promotions to encourage purchases, and enhance product recommendations.
- **Balikpapan and Denpasar:** Conduct retargeting and A/B testing on ads to improve CTR.

Q1: Which marketing factors or channels contributed most to the decline in Conversion Rate (CR)?

2 Deep Analysis – Product



Location	Reach Rate	Engage-ment	CTR	Impres-sion	Product Page View	Purchase	AOV	Conver. Rate
Shoes	0.45	24.27	18.14	12.74K	5.02	525	59.83K	1.27
Pants	0.45	24.60	19.40	12.23K	4.90	497	56.64K	1.14
Accessory	0.43	24.35	17.84	11.98K	4.99	492	56.07K	1.13
Shirt	0.46	24.69	18.84	12.00K	4.93	486	55.38K	1.24

- **Reach Rate:** Accessories have the lowest average (0.43).
- **Engagement:** Shoes have the lowest average (24.27).
- **CTR:** Accessories (17.84%) and Shoes (18.14%) show the lowest averages.
- **Impression:** Accessories (11.98K) and Shirts (12.00K) are the lowest.
- **Product Page View:** Shirts and Pants have the lowest average (4.9).
- **Purchase:** Shirts have the lowest purchase count (486).
- **AOV:** Shirts have the lowest average order value (55.38K).
- **Conversion Rate:** Pants (1.14%) and Accessories (1.13%) show the lowest averages.

Summary

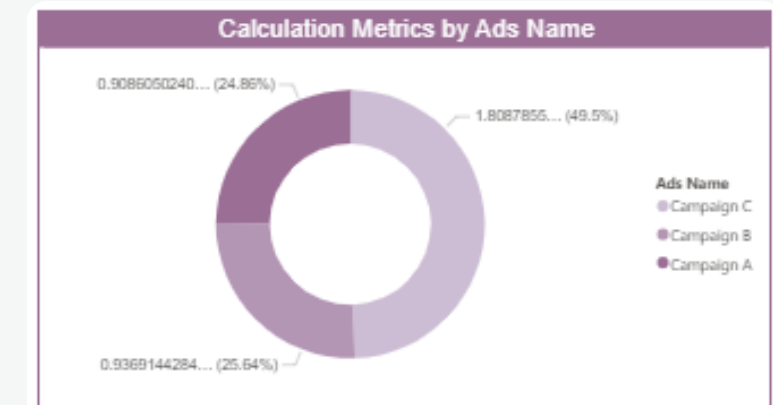
Several product categories show low performance across multiple metrics. Accessories have low reach rate and CTR, Shoes have low engagement and CTR, Shirts have low impressions, product page views, purchases, and AOV, while Pants and Accessories have low conversion rates.

Recommendation

- **Accessories:** Run awareness campaigns to increase visibility, implement bundling or special discounts to attract initial interest, and improve CTR through retargeting and more compelling ad creatives.
- **Shirts:** Launch awareness campaigns and discount-based promotions to boost traffic, optimize product pages (images, description, ratings, trust elements), apply sales strategies such as upselling, cross-selling, or limited-time promotions to drive conversion.
- **Shoes:** Increase engagement through interactive content, user reviews, and loyalty programs, as well as retargeting and creative ad variations to reach more qualified audiences.
- **Pants:** Focus on product page optimization to address low conversion and apply sales-driving strategies such as limited-time offers, upselling, or product recommendations.

Q1: Which marketing factors or channels contributed most to the decline in Conversion Rate (CR)?

3 Deep Analysis – Ads



Accessory

- **CTR:** Campaign A (32.08%) , Campaign B (36.4%), dan Campaign C (31.52%)
- **Rate Reach:** Campaign A (33.94%) , Campaign B (33.41%), dan Campaign C (32.64%)

Pants

- **CTR:** Campaign A (34.91%) , Campaign B (34.15%), dan Campaign C (30.94%)
- **Rate Reach:** Campaign A (35.99%) , Campaign B (33.41%), dan Campaign C (30.6%)

Shirt

- **CTR:** Campaign A (34.92%) , Campaign B (31.48%), dan Campaign C (33.61%)
- **Rate Reach:** Campaign A (33.06%) , Campaign B (34.3%), dan Campaign C (32.63%)

Shoes

- **CTR:** Campaign A (32.5%) , Campaign B (33.54%), dan Campaign C (33.96%)
- **Rate Reach:** Campaign A (32.89%) , Campaign B (33.57%), dan Campaign C (33.54%)

Summary

Overall, Campaigns B and C show lower CTR and Reach Rate compared to Campaign A in several categories. Campaign C performs the lowest in Accessories and Pants, while Campaign B underperforms in Shirts. Meanwhile, Shoes remain stable, indicating consistent ad performance.

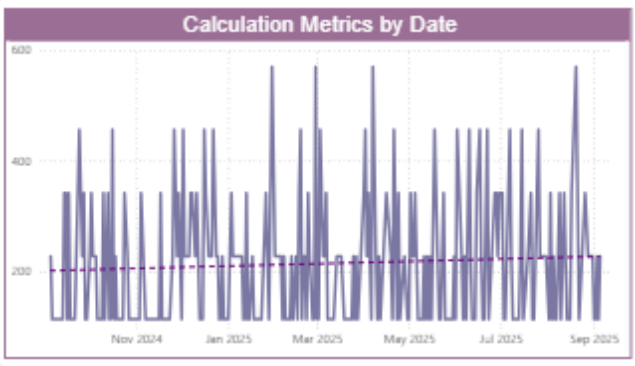
Recommendation

- **Campaign C (Accessories & Pants):** Implement bundling offers or flash sales to boost conversions.
- **Campaign B (Shirts):** Strengthen targeting and enhance creative content.
- **Campaign Shoes:** Maintain the current strategy and focus on customer loyalty.
- **Low CTR:** Use exclusive promotions or apply retargeting strategies.

Q2: How does user behavior on the website and social media influence their purchase decisions?

Q1: How does user behavior on the website and social media influence their purchase decisions?

4 Deep Analysis – Trend Line



Location	CTR	Product Page View	Purchase
Balikpapan	Decreasing	Increasing	Flat
Bandung	Flat	Increasing	Flat
Denpasar	Increasing	Flat	Increasing
Jakarta	Increasing	Increasing	Decreasing
Makassar	Increasing	Decreasing	Decreasing
Medan	Increasing	Decreasing	Flat

Location	CTR	Product Page View	Purchase
Palembang	Decreasing	Increasing	Increasing
Semarang	Decreasing	Increasing	Increasing
Surabaya	Flat	Flat	Decreasing
Yogyakarta	Increasing	Decreasing	Decreasing

Summary

Based on the data by location, the trends vary across regions. Denpasar shows positive performance with increases in CTR and Purchases. In Jakarta, clicks and page views are rising, but Purchases are declining, indicating possible issues during checkout or pricing. Balikpapan and Bandung show high interest, yet Purchases remain stagnant. Makassar and Medan see an increase in clicks but a decrease in page views. Palembang and Semarang experience rising Purchases despite lower CTR, while Surabaya and Yogyakarta show a downward trend in Purchases.

Recommendation

- **Balikpapan, Bandung, Makassar, Medan:** Improve conversion by optimizing the checkout process, offering product bundles, running limited-time promotions, and enhancing product page quality.
- **Jakarta:** Review pricing, implement strategic discounts or time-limited promotions, and improve the user experience on the checkout page to turn clicks into purchases.

Q3: What strategies or optimizations can be implemented to increase the CR in the next three months?

MAIN CHALLENGES ASSOCIATED WITH
ERGONOMICS AND LIVING CONDITIONS



69%
I don't have a
chair I can work
at a time



35%
Too many people
in my household
need to work or
study remotely at
the same time



25%
I don't have a
desk or table
I can work at



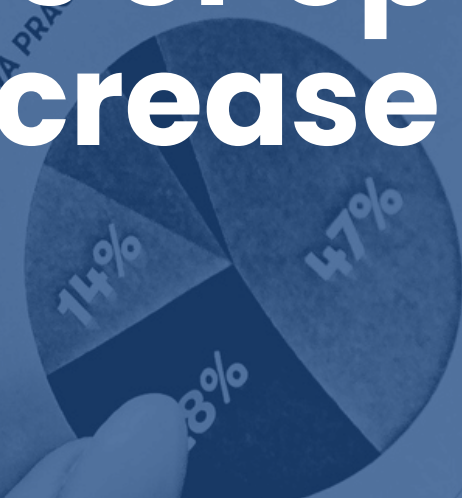
21%
I don't have a
room suitable for
remote work



13%
I don't have a
quiet space to
work

THE FUTURE OF REMOTE WORK:
EMPLOYER AND EMPLOYER PERSPECTIVES

PERSPEKTYWA PRACODAWNIKA



47%
tak, przynajmniej raz w tygodniu

28%
tak, przynajmniej 2 dni w tygodniu

14%
nie chcę pracować zdalnie

11%
tak, więcej niż 3 dni w tygodniu

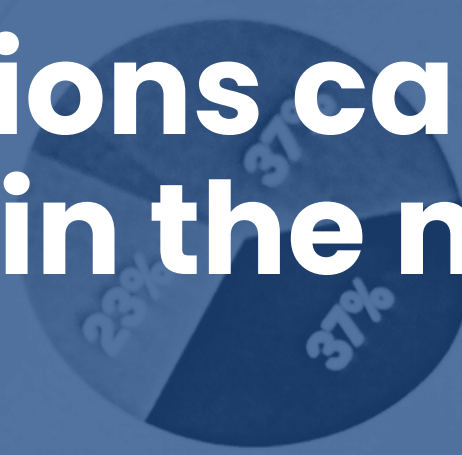
PERSPEKTYWA PRACODAWCY

37%
tak, ale na krótki okres

37%
tak, ale pod warunkiem zachowania
odpowiednich warunków w domu

23%
tak

4%
nie



Q3: What strategies or optimizations can be implemented to increase the CR in the next three months?

Target: Increase the conversion rate by +3% (e.g., from 2% to 5%)

The approach combines technical funnel improvements, segmented marketing campaigns, and A/B testing experiments, with strategies prioritized based on effort versus impact.

Priority Actions

1 Product Page Optimization (+1.0% CR)

Focus on **Shirt, Pants, and Accessory** products. Improvements include **larger product photos, bullet-point benefits, customer reviews, stock/limited-time urgency cues, clear purchase buttons, and cross-sell modules. Conduct A/B testing with trust badges and one-click buy** to measure their effect on conversion from product page to checkout.

2 Checkout & Friction Removal (+0.8% CR)

Focus on **Jakarta**, where clicks are high but purchases are declining. Improvements include **simplifying forms, enabling guest checkout, adding a progress bar, auto-fill features, improved error messages, and real-time validation. Run A/B tests comparing the old checkout flow vs the optimized version.** *Key metrics: checkout conversion and cart abandonment rate.*

3 Retargeting & Creative Ads (+0.5% CR)

Focus on **Campaign B (Shirt) and Campaign C (Accessory & Pants)**. Use **dynamic retargeting, creative testing (video vs carousel), tailored CTAs, and email nurture for abandoned carts.** *Key metric: conversion rate from retargeted traffic vs control.*

4 Bundling / Limited-Time Promos & Flash Sales (+0.4% CR)

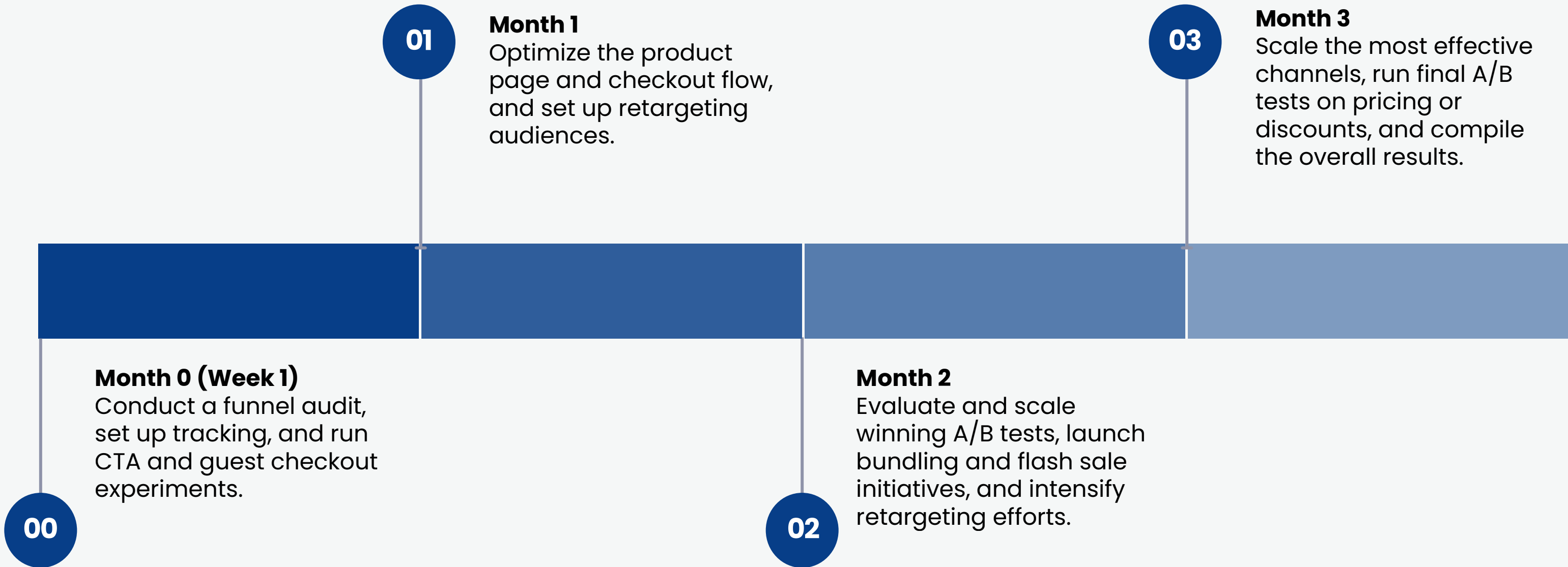
Apply bundling for Accessories and Shirts, and run flash sales in Medan, Makassar, and Denpasar. This strategy increases AOV while simultaneously boosting conversions by enhancing perceived value.

5 Content & Social Engagement (+0.3% CR)

Focus on Denpasar and the **Shoes** category. **Create interactive content such as user-created posts and review videos, and offer loyalty perks.** These efforts are expected to increase engagement, improve click-through rates, and ultimately contribute to higher conversions.

Q3: What strategies or optimizations can be implemented to increase the CR in the next three months?

3-Month Plan



Q3: What strategies or optimizations can be implemented to increase the CR in the next three months?

Experiment & Measurement

Hypothesis: Adding reviews and a trust badge will increase the conversion rate from the product page to checkout by 20%.

- **Unit:** Organic visitors and paid traffic
- **Metric:** Conversion rate from the product page to checkout.
- **Duration:** At least 2 weeks, with a stop rule of $p < 0.05$ and a minimum of 7 days to account for weekday and weekend cycles.
- **Funnel Metrics:** impressions → CTR → page view → add to cart → checkout → purchase.
- **Attribution:** measured using UTM parameters combined with a last-non-direct attribution model, along with a pre/post cohort analysis

Technical Quick Wins

1 Improve Page Speed to Under 3 Seconds

- Ensure the website or product page loads quickly, especially on mobile.
- Slow pages cause visitors to leave the site, which leads to lower conversion rates.

2 Optimize Mobile UX (CTA Thumb Reach, Sticky Add-to-Cart)

- The “Buy” button or any CTA should be easy to reach with the thumb on mobile screens.
- “Sticky add-to-cart” means the purchase button stays visible even while scrolling down the page.

3 Enable One-Click Guest Checkout

- Visitors should be able to purchase without creating an account.
- This reduces checkout friction and increases the number of completed purchases.

Technical Quick Wins

4 Set Up an Abandoned Cart Email Sequence

- Send automated emails when visitors add a product to the cart but do not complete the purchase.
- Emails can be sent after 1 hour, 24 hours, and 72 hours to remind them to finish the checkout.

5 Retargeting Dynamic Ads

- Ads that reappear for visitors who viewed a product but did not buy.
- These ads can show the same product or similar recommendations to increase the chance of conversion.

6 Tweak Creatives for Campaigns B & C

- Adjust ad content (images, videos, CTAs) or audience targeting to improve CTR and conversion rate.

Q3: What strategies or optimizations can be implemented to increase the CR in the next three months?

Monitoring & Interim Targets

Monitor the weekly delta in conversion rate both overall and by city. The interim targets are: +1% by the end of Month 1, +2% by Month 2, and +3% by Month 3. If the targets are not met, scale up the channels that deliver the highest lift.

Specific Recommendations

- **Medan & Makassar:** Implement flash sales and retargeting campaigns.
- **Denpasar:** Use interactive content and loyalty initiatives.
- **Jakarta:** Conduct a pricing and checkout audit.
- **Accessories & Shirts:** Focus on awareness campaigns, bundling offers, and product page optimization.
- **Campaign C:** Bundle Accessories & Pants and measure the post-promo conversion rate.

Thank You!



Github



Power BI